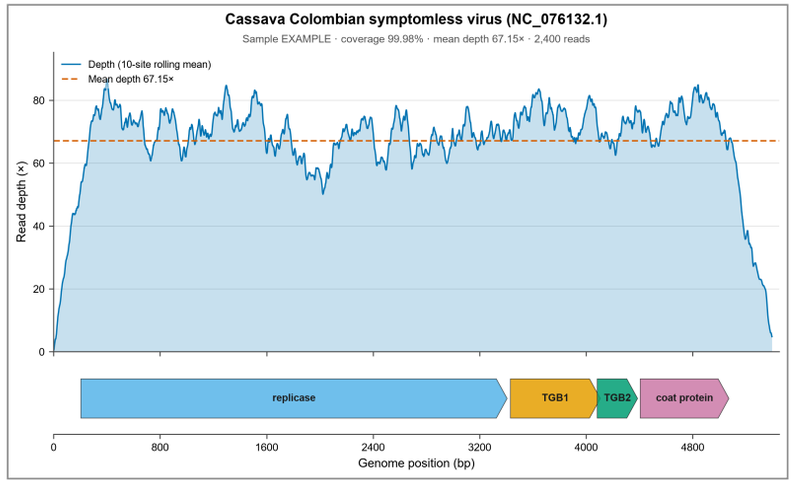
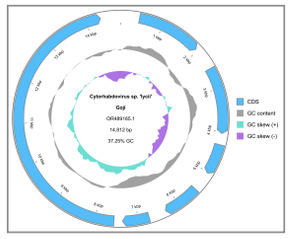
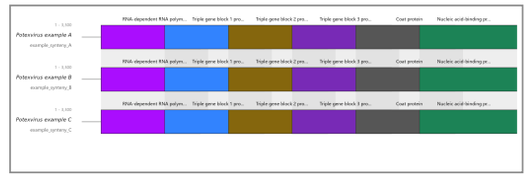
- **VirusPlatform**
  - **Data acquisition**
    - Public data search (SRA + GSA)
    - Bulk download (aria2c + sracha)
    - Genome / host genome download (NCBI datasets)
    - Open-Virome public viromes
    - Sample virome prescreening
    - Local database construction
  - **Virus Detection & Identification**
    - **Sample processing**
      - Sample creation & batch import
      - One-click preprocessing
      - Read QC (fastp)
      - Format conversion (fq2fa)
      - Host read removal (kunpeng)
    - **Quantification & consensus**
      - Known virus ID & quantification (salmon)
      - Coverage depth & BAM
      - Consensus sequence calling
      - Variant analysis
    - **Identification & assembly**
      - One-click pipeline (stage 2-4)
      - Metaviral assembly (SPAdes)
      - Candidate validation (BLASTx / BFVD / viroid)
      - Assembly re-identification
      - Host prediction (ICTV cascade)
  - **Annotation & genomics**
    - **Genome annotation**
      - ORF prediction (pyrodigal)
      - Functional annotation (DIAMOND / Pfam / BFVD)
      - Homology search (BLASTN / BLASTX)
      - Genome maps (gbdraw / DFV)
    - **Comparative genomics**
      - Reference acquisition & CDS/PEP export
      - Multiple alignment (MAFFT + trimAl)
      - Tree inference (RAxML-NG / IQ-TREE / FastTree)
      - SDT pairwise identity (exact engine)
  - **Evolution & Phylodynamics**
    - Metadata governance & alignment QC
    - Temporal signal & dating
    - ML molecular clock (TreeTime)
    - Skyline & Bayesian skyline plot
    - RTT / DRT regression
    - Phylogeography (mugration + migration GIF)
    - Recombination (RDP5)
    - BEAST interface (Markov jump / MCC)
    - One-click dating - geography - tracing
  - **ay design**
    - Primer design (primer3 + thermodynamics)
    - CRISPR detection assay design
    - dsRNA design pipeline (off-target scoring)
    - miRNA target prediction (5 engines)
  - **acing & submission**
    - LOGAN tracing (23.4M public SRA runs)
    - NCBI submission (GenBank / BioSample / MIUVIG)
    - AI assistant (interpret / retrieval)



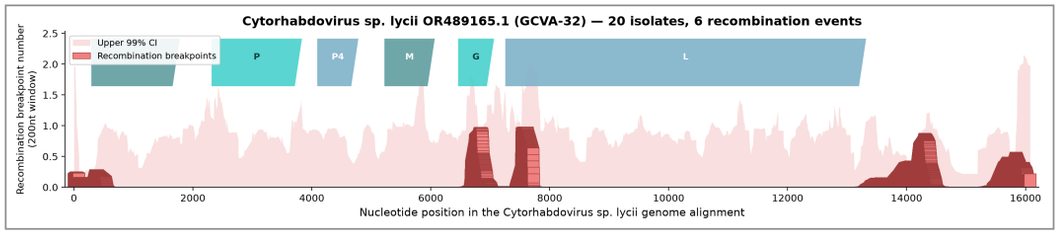
Depth plot (kvsuite)



GCVA genome map (gbdraw)



Synteny comparison



GCVA recombination (RDP5)